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Integrated assemblies and methods of forming integrated assemblies

Abstract

Some embodiments include an integrated assembly having a channel-material-pillar extending vertically through a stack of alternating conductive levels and insulative levels. The channel-material-pillar includes a first semiconductor material. A second semiconductor material is directly against an upper region of the channel-material-pillar. The second semiconductor material has a higher dopant concentration than the first semiconductor material and joins to the first semiconductor along an abrupt interfacial region such that there is little to no mixing of dopant from the second semiconductor material into the first semiconductor material. Some embodiments include methods of forming integrated assemblies.

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Background/Summary

TECHNICAL FIELD

(1) Methods of forming integrated assemblies (e.g., integrated memory devices). Integrated assemblies.

BACKGROUND

(2) Memory provides data storage for electronic systems. Flash memory is one type of memory, and has numerous uses in modern computers and devices. For instance, modern personal computers may have BIOS stored on a flash memory chip. As another example, it is becoming increasingly common for computers and other devices to utilize flash memory in solid state drives to replace conventional hard drives. As yet another example, flash memory is popular in wireless electronic devices because it enables manufacturers to support new communication protocols as they become standardized, and to provide the ability to remotely upgrade the devices for enhanced features.

(3) NAND may be a basic architecture of flash memory, and may be configured to comprise vertically-stacked memory cells.

(4) Before describing NAND specifically, it may be helpful to more generally describe the relationship of a memory array within an integrated arrangement. FIG. **1** shows a block diagram of a prior art device **1000** which includes a memory array **1002** having a plurality of memory cells **1003** arranged in rows and columns along with access lines **1004** (e.g., wordlines to conduct signals WL**0** through WLm) and first data lines **1006** (e.g., bitlines to conduct signals BL**0** through BLn). Access lines **1004** and first data lines **1006** may be used to transfer information to and from the memory cells **1003**. A row decoder **1007** and a column decoder **1008** decode address signals A0 through AX on address lines **1009** to determine which ones of the memory cells **1003** are to be accessed. A sense amplifier circuit **1015** operates to determine the values of information read from the memory cells **1003**. An I/O circuit **1017** transfers values of information between the memory array **1002** and input/output (I/O) lines **1005**. Signals DQ**0** through DQN on the I/O lines **1005** can represent values of information read from or to be written into the memory cells **1003**. Other devices can communicate with the device **1000** through the I/O lines **1005**, the address lines **1009**, or the control lines **1020**. A memory control unit **1018** is used to control memory operations to be performed on the memory cells **1003**, and utilizes signals on the control lines **1020**. The device **1000** can receive supply voltage signals Vcc and Vss on a first supply line **1030** and a second supply line **1032**, respectively. The device **1000** includes a select circuit **1040** and an input/output (I/O) circuit **1017**. The select circuit **1040** can respond, via the I/O circuit **1017**, to signals CSEL**1** through CSELn to select signals on the first data lines **1006** and the second data lines **1013** that can represent the values of information to be read from or to be programmed into the memory cells **1003**. The column decoder **1008** can selectively activate the CSEL**1** through CSELn signals based on the A0 through AX address signals on the address lines **1009**. The select circuit **1040** can select the signals on the first data lines **1006** and the second data lines **1013** to provide communication between the memory array **1002** and the I/O circuit **1017** during read and programming operations.
(5) The memory array **1002** of FIG. **1** may be a NAND memory array, and FIG. **2** shows a schematic diagram of a three-dimensional NAND memory device **200** which may be utilized for the memory array **1002** of FIG. **1**. The device **200** comprises a plurality of strings of charge-storage devices. In a first direction (Z-Z′), each string of charge-storage devices may comprise, for example, thirty-two charge-storage devices stacked over one another with each charge-storage device corresponding to one of, for example, thirty-two tiers (e.g., Tier0-Tier31). The charge-storage devices of a respective string may share a common channel region, such as one formed in a respective pillar of semiconductor material (e.g., polysilicon) about which the string of charge-storage devices is formed. In a second direction (X-X′), each first group of, for example, sixteen first groups of the plurality of strings may comprise, for example, eight strings sharing a plurality (e.g., thirty-two) of access lines (i.e., “global control gate (CG) lines”, also known as wordlines, WLs). Each of the access lines may couple the charge-storage devices within a tier. The charge-storage devices coupled by the same access line (and thus corresponding to the same tier) may be logically grouped into, for example, two pages, such as P0/P32, P1/P33, P2/P34 and so on, when each charge-storage device comprises a cell capable of storing two bits of information. In a third direction (Y-Y′), each second group of, for example, eight second groups of the plurality of strings, may comprise sixteen strings coupled by a corresponding one of eight data lines. The size of a memory block may comprise 1,024 pages and total about 16 MB (e.g., 16 WLs×32 tiers×2 bits=1,024 pages/block, block size=1,024 pages×16 KB/page=16 MB). The number of the strings, tiers, access lines, data lines, first groups, second groups and/or pages may be greater or smaller than those shown in FIG. **2**.
(6) FIG. **3** shows a cross-sectional view of a memory block **300** of the 3D NAND memory device **200** of FIG. **2** in an X-X′ direction, including fifteen strings of charge-storage devices in one of the sixteen first groups of strings described with respect to FIG. **2**. The plurality of strings of the memory block **300** may be grouped into a plurality of subsets **310**, **320**, **330** (e.g., tile columns), such as tile column.sub.l, tile column.sub.j and tile column.sub.K, with each subset (e.g., tile

column) comprising a “partial block” (sub-block) of the memory block **300**. A global drain-side select gate (SGD) line **340** may be coupled to the SGDs of the plurality of strings. For example, the global SGD line **340** may be coupled to a plurality (e.g., three) of sub-SGD lines **342**, **344**, **346** with each sub-SGD line corresponding to a respective subset (e.g., tile column), via a corresponding one of a plurality (e.g., three) of sub-SGD drivers **332**, **334**, **336**. Each of the sub-SGD drivers **332**, **334**, **336** may concurrently couple or cut off the SGDs of the strings of a corresponding partial block (e.g., tile column) independently of those of other partial blocks. A global source-side select gate (SGS) line **360** may be coupled to the SGSs of the plurality of strings. For example, the global SGS line **360** may be coupled to a plurality of sub-SGS lines **362**, **364**, **366** with each sub-SGS line corresponding to the respective subset (e.g., tile column), via a corresponding one of a plurality of sub-SGS drivers **322**, **324**, **326**. Each of the sub-SGS drivers **322**, **324**, **326** may concurrently couple or cut off the SGSs of the strings of a corresponding partial block (e.g., tile column) independently of those of other partial blocks. A global access line (e.g., a global CG line) **350** may couple the charge-storage devices corresponding to the respective tier of each of the plurality of strings. Each global CG line (e.g., the global CG line **350**) may be coupled to a plurality of sub-access lines (e.g., sub-CG lines) **352**, **354**, **356** via a corresponding one of a plurality of sub-string drivers **312**, **314** and **316**. Each of the sub-string drivers may concurrently couple or cut off the charge-storage devices corresponding to the respective partial block and/or tier independently of those of other partial blocks and/or other tiers. The charge-storage devices corresponding to the respective subset (e.g., partial block) and the respective tier may comprise a “partial tier” (e.g., a single “tile”) of charge-storage devices. The strings corresponding to the respective subset (e.g., partial block) may be coupled to a corresponding one of sub-sources **372**, **374** and **376** (e.g., “tile source”) with each sub-source being coupled to a respective power source.
(7) The NAND memory device **200** is alternatively described with reference to a schematic illustration of FIG. **4**.
(8) The memory array **200** includes wordlines **202**.sub.1 to **202**.sub.N, and bitlines **228**.sub.1 to **228**.sub.M.
(9) The memory array **200** also includes NAND strings **206**.sub.1 to **206**.sub.M. Each NAND string includes charge-storage transistors **208**.sub.1 to **208**.sub.N. The charge-storage transistors may use floating gate material (e.g., polysilicon) to store charge, or may use charge-trapping material (such as, for example, silicon nitride, metallic nanodots, etc.) to store charge.
(10) The charge-storage transistors **208** are located at intersections of wordlines **202** and strings **206**. The charge-storage transistors **208** represent non-volatile memory cells for storage of data. The charge-storage transistors **208** of each NAND string **206** are connected in series source-to-drain between a source-select-device (e.g., source-side select gate, SGS) **210** and a drain-select device (e.g., drain-side select gate, SGD) **212**. Each source-select-device **210** is located at an intersection of a string **206** and a source-select line **214**, while each drain-select device **212** is located at an intersection of a string **206** and a drain-select line **215**. The select devices **210** and **212** may be any suitable access devices, and are generically illustrated with boxes in FIG. **4**.
(11) A source of each source-select-device **210** is connected to a common source line **216**. The drain of each source-select-device **210** is connected to the source of the first charge-storage transistor **208** of the corresponding NAND string **206**. For example, the drain of source-select-device **210**.sub.1 is connected to the source of charge-storage transistor **208**.sub.1 of the corresponding NAND string **206**.sub.1. The source-select-devices **210** are connected to source-select line **214**.
(12) The drain of each drain-select device **212** is connected to a bitline (i.e., digit line) **228** at a drain contact. For example, the drain of drain-select device **212**.sub.1 is connected to the bitline **228**.sub.1. The source of each drain-select device **212** is connected to the drain of the last charge-storage transistor **208** of the corresponding NAND string **206**. For example, the source of drain-select device **212**.sub.1 is connected to the drain of charge-storage transistor **208**.sub.N of the

corresponding NAND string **206**.sub.1.
(13) The charge-storage transistors **208** include a source **230**, a drain **232**, a charge-storage region **234**, and a control gate **236**. The charge-storage transistors **208** have their control gates **236** coupled to a wordline **202**. A column of the charge-storage transistors **208** are those transistors within a NAND string **206** coupled to a given bitline **228**. A row of the charge-storage transistors **208** are those transistors commonly coupled to a given wordline **202**.
(14) The vertically-stacked memory cells of three-dimensional NAND architecture may be block-erased by generating hole carriers beneath them, and then utilizing an electric field to sweep the hole carriers upwardly along the memory cells. Gating structures of transistors may be utilized to provide gate-induced drain leakage (GIDL) which generates the holes utilized for block-erase of the memory cells. The transistors may be the source-side select (SGS) devices and/or the drain-side select (SGD) devices.
(15) It is desired to develop improved methods of forming integrated memory (e.g., NAND memory). It is also desired to develop improved memory devices.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** shows a block diagram of a prior art memory device having a memory array with memory cells.
(2) FIG. **2** shows a schematic diagram of the prior art memory device of FIG. **1** in the form of a 3D NAND memory device.
(3) FIG. **3** shows a cross-sectional view of the prior art 3D NAND memory device of FIG. **2** in an X-X′ direction.
(4) FIG. **4** is a schematic diagram of a prior art NAND memory array.
(5) FIGS. **5** and **5**A are a diagrammatic cross-sectional side view (FIG. **5**) and a diagrammatic cross-sectional top-down view (FIG. **5**A) of a region of an example integrated assembly comprising example memory cells along an example channel-material-pillar. The top-down view of FIG. **5**A is along the line A-A of FIG. **5**.
(6) FIG. **6** is a diagrammatic cross-sectional side view of a region of an example integrated assembly comprising example memory cells along an example channel-material-pillar. A view along the line A-A of FIG. **6** may be the same as the cross-sectional top-down view of FIG. **5**A.
(7) FIGS. **7**A and **7**B are diagrammatic cross-sectional side views of regions of example integrated assembly showing heavily-doped semiconductor material directly adjacent to less-doped semiconductor material.
(8) FIGS. **8-13** are diagrammatic cross-sectional side views of regions of an example integrated assembly at example sequential process stages of an example method.
DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENTS
(9) Some embodiments include integrated assembly having channel material (lightly-doped or undoped semiconductor material) directly adjacent to heavily-doped semiconductor material, and having a sharp dopant interface along a region where the two materials join to one another. Some embodiments include methods of forming integrated assemblies (e.g., memory devices). Example embodiments are described with reference to FIGS. **5-13**.
(10) Referring to FIG. **5**, a construction (i.e., assembly, architecture, etc.) **10** includes a stack **12** of alternating first and second levels **14** and **16**, with such levels being supported over a source structure **28**.
(11) The first levels **14** comprise materials **18** and **20**, with the material **18** being conductive and the material **20** being insulative. In some embodiments, the conductive material **18** may comprise two or more conductive compositions. For instance, the material **18** may comprise a metal-containing

core and a metal nitride composition peripherally surrounding the core. The core composition may comprise, for example, tungsten, titanium, tantalum, etc. The metal nitride composition may comprise, for example, tungsten nitride, titanium nitride, etc. The insulative material **20** may comprise one or more high-k compositions (e.g., aluminum oxide, zirconium oxide, hafnium oxide, etc.), with the term “high-k” meaning a dielectric constant greater than that of silicon dioxide. In some embodiments, the insulative material **20** may correspond to a dielectric barrier material.
(12) The second levels **16** comprise insulative material **22**. The insulative material **22** may comprise any suitable composition(s), and in some embodiments may comprise, consist essentially of, or consist of silicon dioxide.
(13) The levels **14** and **16** may be of any suitable thicknesses, and may be the same thickness as one another or different thicknesses relative to one another. In some embodiments, the levels **14** and **16** may have vertical thicknesses within a range of from about 10 nanometers (nm) to about 400 nm.
(14) In the illustrated embodiment, an insulative level **15** is over the uppermost conductive level **14** of the stack **12**. The insulative level **15** is vertically thicker than the other insulative levels **16**. In some embodiments, the uppermost insulative level **15** may be at least about twice as thick as the other insulative levels **16**. Although the stack **12** is shown to not include the uppermost insulative level **15**, in other embodiments the stack **12** may be considered to include the level **15** in addition to the levels **14** and **16**.
(15) The insulative level **15** may comprise any suitable composition(s), and in the shown embodiment comprises the same insulative composition **22** as the other insulative levels **16**.
(16) In some embodiments, the stack **12** may be considered to comprise alternating conductive levels **14** and insulative levels **16**. Some of the conductive levels **14** may correspond to a wordline/memory cell levels **24**, and others may correspond to SGD levels **26**. In the shown embodiment, the upper three of the conductive levels **14** are shown to correspond to SGD levels. Generally, one or more of the uppermost levels **14** will correspond to SGD levels. In some embodiments, the number of SGD levels will be within a range of from at least 1 to about 10. If multiple conductive levels are utilized as SGD levels, the conductive levels may be electrically coupled with one another (ganged together) to be incorporated into long-channel SGD devices.
(17) There may be any suitable number of the wordline/memory cell levels **24**. For instance, in some embodiments there may be 8 memory cell levels, 16 memory cell levels, 32 memory cell levels, 64 memory cell levels, 512 memory cell levels, 1024 memory cell levels, etc. The vertical stack **12** is diagrammatically indicated to extend downwardly beyond the illustrated region of the stack to indicate that there may be more vertically-stacked levels than those specifically illustrated in the diagram of FIG. **5**.
(18) The source structure **28** may comprise any suitable composition(s), and in some embodiments may comprise conductively-doped semiconductor material (e.g., conductively-doped silicon) over metal-containing material (with example metal-containing materials including one or more of tungsten, tungsten silicide, titanium, etc.).
(19) The source structure **28** is shown to be supported by a base **30**. The base **30** may comprise semiconductor material; and may, for example, comprise, consist essentially of, or consist of monocrystalline silicon. The base **30** may be referred to as a semiconductor substrate. The term “semiconductor substrate” means any construction comprising semiconductive material, including, but not limited to, bulk semiconductive materials such as a semiconductive wafer (either alone or in assemblies comprising other materials), and semiconductive material layers (either alone or in assemblies comprising other materials). The term “substrate” refers to any supporting structure, including, but not limited to, the semiconductor substrates described above. In some applications, the base **30** may correspond to a semiconductor substrate containing one or more materials associated with integrated circuit fabrication. Such materials may include, for example, one or more of refractory metal materials, barrier materials, diffusion materials, insulator materials, etc.

(20) A gap is provided between the base **30** and the source structure **28** to indicate that additional materials, components, etc., may be provided between the base **30** and the source structure **28** in some embodiments.
(21) The base **30** is shown to have a horizontally-extending upper surface **31**.
(22) A cell-material-pillar **32** is shown to extend vertically through the stack **12**. The cell-material-pillar **32** may be considered to be representative of a large number of substantially identical cell-material-pillars, with the term "substantially identical" meaning identical to within reasonable tolerances of fabrication and measurement. The pillars **32** may be configured in a tightly-packed arrangement, such as, for example, a hexagonal close packed (HCP) arrangement. There may be hundreds, thousands, millions, hundreds of thousands, etc., of the cell-material-pillars **32** extending through the stack **12**.
(23) The vertically-extending pillar **32** may extend at any suitable angle relative to the horizontally-extending upper surface **31** of the base **30**. In some embodiments, the pillar **32** may be orthogonal, or at least substantially orthogonal, relative to the horizontally-extending surface **31**, with the term "substantially orthogonal" meaning orthogonal to within reasonable tolerances of fabrication and measurement. In some embodiments, the pillar **32** may to extend within about ±15° of orthogonal relative to the horizontally-extending surface **31** of the base **30**.
(24) The pillar **32** comprises an insulative core material **34**, a channel material **36**, a tunneling material **38**, a charge-storage material **40** and a charge-blocking material **42**.
(25) The channel material **36** is shown with stippling to assist the reader in identifying the channel material. The channel material **36** comprises semiconductor material. The semiconductor material may comprise any suitable composition(s); and in some embodiments may comprise, consist essentially of, or consist of one or more of silicon, germanium, III/V semiconductor material (e.g., gallium phosphide), semiconductor oxide, etc.; with the term III/V semiconductor material referring to semiconductor materials comprising elements selected from groups III and V of the periodic table (with groups III and V being old nomenclature, and now being referred to as groups **13** and **15**). In some embodiments, the semiconductor material may comprise, consist essentially of, or consist of appropriately-doped silicon.
(26) In some embodiments, the channel material **36** may comprise undoped semiconductor material, such as, for example, undoped silicon. The term "undoped" doesn't necessarily mean that there is absolutely no dopant present within the semiconductor material, but rather means that any dopant within such semiconductor material is present to an amount generally understood to be insignificant. For instance, undoped silicon may be understood to comprise a dopant concentration of less than about 10.sup.16 atoms/cm.sup.3, less than about 10.sup.15 atoms/cm.sup.3, etc., depending on the context. In some embodiments, the channel material **36** may comprise, consist essentially of, or consist of silicon. In some embodiments, the channel-material **36** may comprise silicon which is lightly-doped with appropriate n-type and/or p-type dopant (e.g., one or more of phosphorus, arsenic, boron, etc.), with a maximum total concentration of dopant within the channel material being less than or equal to about 10.sup.18 atoms/cm.sup.3.
(27) The semiconductor material within the channel material **36** may be referred to as a first semiconductor material to distinguish it from other semiconductor materials present within the integrated assembly **10**.
(28) The channel material **36** may be considered to be configured as a channel-material-pillar **44**. The pillar **44** may be configured in a hollow-pillar-configuration comprising a cylindrical wall **45** laterally surrounding a hollow **47**, as shown. FIG. **5**A shows a cross-section along the line A-A of FIG. **5**, and shows the illustrated region of the channel-material-pillar **44** configured as a ring (annulus, donut-shape, annular-ring, etc.) along the top-down cross-section. Such ring may be considered to comprise the cylindrical wall **45** laterally surrounding the hollow **47**. The cylindrical wall has an inner surface **49** along the hollow **47** and directly contacting the insulative material **34**, and has an outer surface **51** directly contacting the tunneling material **38**. The cylindrical wall has a

lateral thickness T between the inner and outer walls **49** and **51**. Such lateral thickness may be of any suitable dimension, and in some embodiments may be of a dimension which is within a range of from about 4 nm to about 30 nm.

(29) Although the channel-material-pillar **44** is shown to be configured as a “hollow” channel configuration, in other embodiments the pillar **44** may be configured as a solid pillar rather than as a hollow pillar.

(30) The tunneling material **38** (also referred to as gate dielectric material) may comprise any suitable composition(s), and in some embodiments may comprise one or more of silicon dioxide, aluminum oxide, hafnium oxide, zirconium oxide, etc.

(31) The charge-storage material **40** may comprise any suitable composition(s), and in some embodiments may comprise floating gate material (e.g., polysilicon) or charge-trapping material (e.g., one or more of silicon nitride, silicon oxynitride, conductive nanodots, etc.).

(32) The charge-blocking material **42** may comprise any suitable composition(s), and in some embodiments may comprise one or more of silicon dioxide, aluminum oxide, hafnium oxide, zirconium oxide, etc.

(33) An SGS device **46** is shown to be associated with a lower region of the channel-material pillar **44** in the side view of FIG. **5**. Also, the channel-material-pillar **44** is shown to be electrically coupled with the source structure **28**. Memory cells **48** are along the memory cell levels **24**, and SGD devices **50** are along the SGD levels **26**.

(34) Each of the memory cells **48** comprises a region of the semiconductor material (channel material) **36**, and comprises regions (control gate regions) of the conductive levels **14**. The regions of the conductive levels which are not comprised by the memory cells **48** may be considered to be wordline regions (routing regions) which couple the control gate regions with driver circuitry and/or other suitable circuitry. The memory cells **48** comprise the cell materials **38**, **40**, **42** and **20**, in addition to comprising the channel material **36**.

(35) The memory cells **48** are vertically stacked one atop another.

(36) In operation, the charge-storage material **40** may be configured to store information in the memory cells **50**. The value (with the term “value” representing one bit or multiple bits) of information stored in an individual memory cell may be based on the amount of charge (e.g., the number of electrons) stored in a charge-storage region. The amount of charge within an individual charge-storage region may be controlled (e.g., increased or decreased), at least in part, based on the value of voltage applied to a gate, and/or based on the value of voltage applied to the channel.

(37) The tunneling material **38** may be configured to allow desired tunneling (e.g., transportation) of charge (e.g., electrons) between the charge-storage material **40** and the channel material **36**. The tunneling material **38** may be configured (i.e., engineered) to achieve a selected criterion, such as, for example, but not limited to, an equivalent oxide thickness (EOT). The EOT quantifies the electrical properties of the tunneling region (e.g., capacitance) in terms of a representative physical thickness. For example, EOT may be defined as the thickness of a theoretical silicon dioxide layer that would be required to have the same capacitance density as a given dielectric, ignoring leakage current and reliability considerations.

(38) The charge-blocking material **42** is adjacent to the charge-storage material **40**, and may provide a mechanism to block charge from flowing from the charge-storage material **40** to the gates along conductive levels **14**.

(39) The dielectric barrier material **20** is provided between the charge-blocking material **42** and the associated gates along the conductive levels **14**, and may be utilized to inhibit back-tunneling of electrons from the gates toward the charge-storage material **40**.

(40) A second semiconductor material **52** is over the cell-material-pillar **32**, and directly contacts an upper region of the channel-material-pillar **44**. In the illustrated embodiment, the second semiconductor material **52** directly contacts an upper surface **53** of the channel-material-pillar **44**. In some embodiments, the second semiconductor material **52** may be considered to be configured

as a semiconductor-material-plug **66**.

(41) In some embodiments, the first and second semiconductor materials **36** and **52** comprise a same composition as one another. For instance, the first and second semiconductor materials **36** and **52** may both comprise silicon. The silicon within the second semiconductor material **52** may be in any suitable phase, and in some embodiments may be in one or both of an amorphous phase and a polycrystalline phase.

(42) The second semiconductor material **52** has a higher dopant concentration than the first semiconductor material **36** (i.e., the channel material). In some embodiments, the second semiconductor material **52** may comprise silicon having a total dopant concentration of one or more suitable n-type and/or p-type dopants (e.g., phosphorus, boron, arsenic, etc.) of greater than or equal to about 10.sup.20 atoms/cm.sup.3, greater than or equal to about 10.sup.21 atoms/cm.sup.3, etc.

(43) The first and second semiconductor materials **36** and **52** join to one another along an interfacial region **54**. In the illustrated embodiment of FIG. **5**, such interfacial region is coextensive with the upper surface **53** of the channel-material-pillar **44**.

(44) The interfacial region **54** is above the SGD levels **26**.

(45) In operation, the memory cells **48** may be block-erased utilizing GIDL established by the SGS and SGD devices (**46** and **50**). A difficulty which may be encountered in conventional NAND-memory configurations is that there may be a dopant gradient along an upper region of the channel material within the channel-material pillars, with such gradient being a suboptimal dopant profile for GIDL generation. Instead, it is desired to have a dopant profile corresponding to a uniform and consistent dopant concentration throughout the entirety of the channel material **36**, and to have an abrupt interface between the low-dopant-concentration-semiconductor-material **36** and the high-dopant-concentration-semiconductor-material **52**. The structure of FIG. **5** may have the desired dopant profile. The concept of an abrupt interface between a low-dopant-concentration-semiconductor-material and a high-dopant-semiconductor-material is described in more detail below with reference to FIGS. **7**A and **7**B.

(46) Referring still to FIG. **5**, a first conductive interconnect **54** is provided over the semiconductor material **52** and is electrically coupled with the semiconductor material **52**. A second conductive interconnect **56** is provided over the first conductive interconnect **54** and is electrically coupled with the first conductive interconnect **54**. A bitline (digit line, sense line, etc.) **58** is provided over the interconnect **56** and is electrically coupled with the interconnect **56**. Accordingly, the bitline **58** is electrically coupled with the channel material **36** through the semiconductor material **52** and the interconnects **54** and **56**.

(47) The bitline **58** may extend in and out of the page relative to the cross-sectional view of FIG. **5**.

(48) The interconnects **54** and **56** comprise conductive materials **60** and **62**, respectively. The materials **60** and **62** may comprise any suitable electrically conductive composition(s); such as, for example, one or more of various metals (e.g., titanium, tungsten, cobalt, nickel, platinum, ruthenium, etc.), metal-containing compositions (e.g., metal silicide, metal nitride, metal carbide, etc.), and/or conductively-doped semiconductor materials (e.g., conductively-doped silicon, conductively-doped germanium, etc.). In some embodiments, the materials **60** and **62** may both be metal-containing materials (e.g., may comprise one or more of tungsten, titanium, tungsten nitride, titanium nitride, etc.). The materials **60** and **62** may be compositionally the same as one another, or may be compositionally different relative to one another.

(49) FIG. **6** shows another embodiment of an integrated assembly having the second semiconductor material **52** over the channel material **36** of the channel-material-pillar **44**. The top-down cross-sectional view along the line A-A of FIG. **6** will be identical to FIG. **5**A.

(50) FIG. **6** shows the insulative material (dielectric material) **34** filling only a lower region **64** of the hollow **47**. The semiconductor material **52** of the semiconductor-material-plug **66** extends into an upper region **68** of the hollow.

(51) The interfacial region **54** between the first and second semiconductor materials **36** and **52** includes a portion of the upper surface **53**, and includes a portion of the inner surface **49** of the cylindrical wall **45** within the upper region **68** of the hollow **47**.
(52) In the shown embodiment, the semiconductor-material-plug **66** is configured to include tapered sidewalls **67** which extend to an uppermost surface **64**. In some embodiments, the channel-material-pillar **44** may be considered to have a first lateral width W.sub.1 along the cross-section of FIG. **6**, and the uppermost surface **64** of the plug **66** may be considered to have a second lateral width W.sub.2 along the cross-section, with the second lateral width being greater than the first lateral width. In some embodiments, the second lateral width may be large enough that the interconnect **54** of FIG. **5** may be omitted, and instead only the interconnect **56** extends between the plug **66** and the bitline **58** (as shown).
(53) The plug **66** is shown to be directly against a portion of the uppermost surface **53** of the channel-material-pillar **44**, and the material **22** is directly against another portion of the uppermost surface **53** of the channel-material-pillar. In other embodiments, the plug **66** may extend entirely across the uppermost surface **53** of the channel-material-pillar or may not extend across any of the uppermost surface **53** of the channel-material-pillar.
(54) The upper region **68** of the hollow **47** may have any suitable vertical dimension D, and in some embodiments may have a vertical dimension of at least about 20 nm, at least about 40 nm, etc.
(55) In some embodiments, an advantage of the assemblies of FIGS. **5** and **6** is that there will be little, if any, intermixing of dopant from the second semiconductor material **52** into the first semiconductor material **36** (i.e., from the heavily-doped conductive interconnect material **52** into the undoped, or lightly-doped, channel material **36**). FIGS. **7**A and **7**B diagrammatically illustrate example embodiments in which there is little or no intermixing of dopant from the heavily-doped material **52** into the channel material **36**.
(56) FIG. **7**A shows a region of the construction of FIG. **6** having the channel material **36** (illustrated with stippling) adjacent to the heavily-doped semiconductor material **52** within the upper portion **68** of the hollow **47**. The tunneling material **38** is shown to be on an opposite side of the channel material **36** from the heavily-doped semiconductor material **52**. The channel material **36** has the lateral thickness T between the materials **52** and **38**, with an example lateral thickness T being within a range of from greater than or equal to about 4 nm to less than or equal to about 30 nm. FIG. **7**A may be considered to illustrate an embodiment in which there is no intermixing of dopant from the heavily-doped material **52** into the channel material **36**. Accordingly, the interfacial region **54** is simply a boundary between the materials **52** and **36**.
(57) In contrast, FIG. **7**B illustrates a configuration similar to that of FIG. **7**A, but in which there is a little mixing of dopant from the material **52** into the material **36**, and accordingly in which the interfacial region **54** extends a distance X into the channel material **36**. The distance X is less than the full lateral thickness T of the cylindrical wall **45** comprising the channel material **36**, and in some embodiments may be less than or equal to about 50% of the lateral thickness T, less than or equal to about 20% of such lateral thickness, less than or equal to about 10% of such lateral thickness, less than or equal to about 5% of such lateral thickness, etc. In some embodiments, any mixing of dopant from the second semiconductor material **52** into the channel material **36** will be less than or equal to about one-third of the lateral thickness T, less than or equal to about one-fourth of such lateral thickness, etc. In some embodiments, any mixing of dopant from the second semiconductor material **52** into the first semiconductor material **36** will extend less than or equal to about 20 nm into the channel material, less than or equal to about 5 nm into the channel material, less than or equal to about 2 nm into the channel material, etc.
(58) It is desired that there be little mixing of dopant from the heavily-doped material **52** into the channel material **36** in order to avoid the problems with GIDL described above as being problematically associated with conventional configurations having suboptimal dopant profiles

along the channel material **36**. It can be particularly desired that any mixing of dopant from the heavily-doped material **52** into the channel material **56** does not extend entirely across the lateral sidewall of the material **36** relative to the embodiment of FIG. **6** in which the material **52** extends into the upper region **68** of the hollow **47**. In contrast, in the embodiment of FIG. **5** in which the second material **52** extends entirely across the upper surface **53** of the channel material **36**, it can simply be desired that there be either no detectable mixing of dopant from the material **52** into the channel material **36**, or that any mixing be limited to a region above the uppermost SGD level **26**, and preferably be limited to extend less than or equal to about 10 nm into the channel material **36**, less than or equal to about 5 nm into the channel material, less than or equal to about 2 nm into the channel material, etc.
(59) The configurations of FIGS. **5** and **6** may be formed with any suitable processing. FIGS. **8-13** describes example processing for forming the configuration of FIG. **6**.
(60) Referring to FIG. **8**, a region of the assembly **10** is shown at an example process stage. The source structure **28** and the base **30** are not shown in FIG. **8** in order to simplify the drawing.
(61) The assembly **10** includes a stack **12** of alternating first and second levels **14** and **16**. The second levels **16** comprise the insulative material **22** described above. The first levels **14** comprise a sacrificial material **70**. The sacrificial material **70** may comprise any suitable composition(s), and in some embodiments may comprise, consist essentially of, or consist of silicon nitride. The material **22** may be referred to as a first insulative material to distinguish it from other insulative materials, and the sacrificial material **70** may be referred to as a first sacrificial material to distinguish it from other sacrificial materials.
(62) A thick layer **72** of the material **22** is formed over the stack **12**. In some embodiments, the thick layer **72** may be considered to be part of the stack **12**.
(63) Cell-material-pillars **32** are formed to extend through the stack **12** and the layer **72**. The cell-material-pillars comprise the channel-material-pillars **44** configured as hollow cylinders. The hollow cylinders have the cylindrical sidewalls **45** surrounding the hollows **47**. The insulative material **34** fills the lower regions **64** of the hollows **47**. In some embodiments, the insulative material **34** may be referred to as a second insulative material.
(64) The cell-material-pillars **32** include regions **74** outwardly of the channel material **36** and laterally surrounding the channel material. The regions **74** may include the tunneling material, charge-storage material and a charge-blocking material described above with reference to FIG. **5**. The tunneling material, charge-storage material and a charge-blocking material may be collectively referred to as cell materials.
(65) Sacrificial material **76** is formed within the upper regions **68** of the hollows **47**. The sacrificial material **76** may be referred to as second sacrificial material. The sacrificial material **76** may comprise any suitable composition(s), and in some embodiments may comprise one or more of silicon nitride, carbon (e.g., amorphous carbon), carbon-doped silicon dioxide, metal, aluminum oxide, etc.
(66) A planarized surface **73** is formed to extend across the layer **72**, the pillars **32** and the sacrificial material **76**. The planarized surface **73** may be formed with any suitable processing, including, for example, chemical-mechanical polishing (CMP). In some embodiments, the layer **72** may be considered to be an uppermost of the insulative levels (second levels) **14**, and accordingly the planarized surface **73** may be considered to extend across such uppermost of the second levels.
(67) Referring to FIG. **9**, additional insulative material **22** is formed over the layer **72** to increase a vertical thickness of the layer **72**, and in some embodiments may be considered to be formed across the planarized surface **73** (FIG. **8**). Although the same material **22** is shown to be formed over the planarized surface **73** as is utilized within the layer **72** of FIG. **8**, in other embodiments a different insulative material may be formed over the planarized surface **73** than is utilized within the layer **72**. The additional material formed over the planarized surface **73** may be referred to as a third insulative material in some embodiments.

(68) In some embodiments, the layer **72** of FIG. **9** may have a thickness which extends over the cell-material-pillars **32** by a distance Y of at least about 100 nm, and in some embodiments such distance Y may be within a range of from about 100 nm to about 1 micron (μm).

(69) The sacrificial material **70** of FIG. **8** is removed and replaced with the conductive material **18** to form the conductive levels **14** described above with reference to FIGS. **5** and **6**. Although the dielectric barrier material **20** is not shown in FIG. **9**, it is to be understood that such material may be formed within the levels **14** in addition to the conductive material **18**. If the dielectric barrier material is within the regions **74** of the cell-material-pillars, then the sacrificial material **70** (FIG. **8**) may be removed and replaced entirely with the conductive material **18**. Alternatively, if it is desired to form the dielectric barrier material along the levels **14**, then the sacrificial material **70** (FIG. **8**) may be removed and replaced with both the conductive material **18** and the dielectric barrier material **20** (to form a configuration analogous to that shown in FIG. **5**).

(70) The sacrificial material **70** (FIG. **8**) may be removed by forming slits (not shown) through the stack **12** to expose regions of the levels **14** and **16**, and then selectively removing the sacrificial material **70** relative to other exposed materials. Subsequently, appropriate materials and/or precursors may be flowed into the slits to form the conductive material **18** (and also possibly the dielectric barrier material **20**) along the levels **14**.

(71) Referring to FIG. **10**, openings **78** are formed to expose the sacrificial material **76**. The openings **78** may have depths corresponding to the dimension Y, and accordingly may have depths within the range of from about 100 nm to about 1 μm. The openings **78** may have bottom widths W.sub.3 along the cross-section of FIG. **10** within a range of from about 50 nm to about 100 nm, and may have top widths W.sub.2 along the cross-section of FIG. **10** within a range of from about 100 nm to about 200 nm.

(72) In the illustrated embodiment, outer edges of the openings **78** land on upper surfaces **53** of the channel material **36**. In other embodiments, the openings may have narrower bottom widths such that the outer edges of the openings land on the sacrificial material **76**, or may have wider bottom widths such that the outer edges of the openings land on the cell materials within the regions **74**, or even land outwardly of the regions **74** and within the insulative material **22** of the layer **72**.

(73) The flexibility associated with the landing regions of the openings **78** may be advantageous during fabrication of an integrated assembly **10** in that such may provide tolerance to compensate for mask misalignment that may occur during such fabrication.

(74) Referring to FIG. **11**, the insulative material **76** (FIG. **10**) is removed to extend the openings **78** to upper surfaces of the insulative material **34** within the hollows **47** defined by the cylindrical walls **45** of the channel-material-pillars **44**.

(75) Referring to FIG. **12**, the heavily-doped semiconductor material **52** is formed within the openings **78** (FIG. **11**), and then a planarized surface **79** is formed to extend across the materials **22** and **52**. The surface **79** may be formed with any suitable processing, including, for example, CMP.

(76) The material **52** of FIG. **12** is configured as conductive plugs **66** analogous to the plug described above with reference to FIG. **6**.

(77) Referring to FIG. **13**, interconnects **56** and bitlines **58** are formed over the conductive plugs **66**. The bitlines **58** are coupled to the channel material **36** of the channel-material-pillars **44** through the interconnects **56** and the conductive plugs **66**.

(78) The configuration of FIG. **13** may be considered to be analogous to that described above with reference to FIG. **6**. The memory cells **48** and SGD devices **50** are associated with the conductive levels **14**, and are along regions of the cell-material-pillars **32**.

(79) In some embodiments, any high-temperature thermal processing (e.g., thermal processing utilizing temperatures in excess of 1000° C.) may be conducted prior to forming the semiconductor material **52**. Accordingly, interfaces between the semiconductor materials **52** and **36** will not be subjected to thermal stresses which may inadvertently cause undesired intermixing of dopant from the material **52** into the material **36**. In some embodiments, it may be desired to anneal the material

52 to activate dopant within such material. Such annealing may be conducted with relatively low-temperature thermal processing (e.g., thermal processing utilizing a maximum temperature of less than or equal to about 600° C.). In some embodiments, the conductive plugs **66** of FIG. **12** may be exposed to suitable low-temperature thermal processing to activate dopant (e.g., phosphorus) within the heavily-doped semiconductor material **52** without substantially intermixing dopant from the material **52** into the channel material **36**.

(80) The assemblies and structures discussed above may be utilized within integrated circuits (with the term “integrated circuit” meaning an electronic circuit supported by a semiconductor substrate); and may be incorporated into electronic systems. Such electronic systems may be used in, for example, memory modules, device drivers, power modules, communication modems, processor modules, and application-specific modules, and may include multilayer, multichip modules. The electronic systems may be any of a broad range of systems, such as, for example, cameras, wireless devices, displays, chip sets, set top boxes, games, lighting, vehicles, clocks, televisions, cell phones, personal computers, automobiles, industrial control systems, aircraft, etc.

(81) Unless specified otherwise, the various materials, substances, compositions, etc. described herein may be formed with any suitable methodologies, either now known or yet to be developed, including, for example, atomic layer deposition (ALD), chemical vapor deposition (CVD), physical vapor deposition (PVD), etc.

(82) The terms “dielectric” and “insulative” may be utilized to describe materials having insulative electrical properties. The terms are considered synonymous in this disclosure. The utilization of the term “dielectric” in some instances, and the term “insulative” (or “electrically insulative”) in other instances, may be to provide language variation within this disclosure to simplify antecedent basis within the claims that follow, and is not utilized to indicate any significant chemical or electrical differences.

(83) The terms “electrically connected” and “electrically coupled” may both be utilized in this disclosure. The terms are considered synonymous. The utilization of one term in some instances and the other in other instances may be to provide language variation within this disclosure to simplify antecedent basis within the claims that follow.

(84) The particular orientation of the various embodiments in the drawings is for illustrative purposes only, and the embodiments may be rotated relative to the shown orientations in some applications. The descriptions provided herein, and the claims that follow, pertain to any structures that have the described relationships between various features, regardless of whether the structures are in the particular orientation of the drawings, or are rotated relative to such orientation.

(85) The cross-sectional views of the accompanying illustrations only show features within the planes of the cross-sections, and do not show materials behind the planes of the cross-sections, unless indicated otherwise, in order to simplify the drawings.

(86) When a structure is referred to above as being “on”, “adjacent” or “against” another structure, it can be directly on the other structure or intervening structures may also be present. In contrast, when a structure is referred to as being “directly on”, “directly adjacent” or “directly against” another structure, there are no intervening structures present. The terms “directly under”, “directly over”, etc., do not indicate direct physical contact (unless expressly stated otherwise), but instead indicate upright alignment.

(87) Structures (e.g., layers, materials, etc.) may be referred to as “extending vertically” to indicate that the structures generally extend upwardly from an underlying base (e.g., substrate). The vertically-extending structures may extend substantially orthogonally relative to an upper surface of the base, or not.

(88) Some embodiments include an integrated assembly having a channel-material-pillar extending vertically through a stack of alternating conductive levels and insulative levels. The channel-material-pillar includes a first semiconductor material. A second semiconductor material is directly against an upper region of the channel-material-pillar. The second semiconductor material has a

higher dopant concentration than the first semiconductor material and joins to the first semiconductor material along an abrupt interfacial region such that there is little or no mixing of dopant from the second semiconductor material into the first semiconductor material.
(89) Some embodiments include an integrated assembly comprising a channel-material-pillar extending vertically through a stack of alternating conductive levels and insulative levels. The channel-material-pillar comprises a first semiconductor material. The channel-material-pillar has a hollow-pillar-configuration comprising a cylindrical wall laterally surrounding a hollow. The cylindrical wall has an inner surface along the hollow, and has a lateral thickness. A dielectric material fills a lower region of the hollow. An upper region of the hollow is above said lower region. A semiconductor-material-plug is over the stack and extends into the upper region of the hollow. The semiconductor-material-plug comprises a second semiconductor material. The second semiconductor material of the semiconductor-material-plug is directly against the inner surface of the cylindrical wall along the upper region of the hollow. The second semiconductor material has a higher dopant concentration than the first semiconductor material. Any intermixing of dopant from the second semiconductor material into the first semiconductor material extends less than the lateral thickness of the cylindrical wall.
(90) Some embodiments include a method of forming an integrated assembly. A stack is formed to comprise alternating first and second levels. The first levels comprise first sacrificial material and the second levels comprise first insulative material. Pillars are formed to extend through the stack. The pillars include cell materials, channel material and second insulative material. The channel material is configured as hollow cylinders having cylindrical sidewalls surrounding hollows. The second insulative material fills lower regions of the hollows. The cell materials laterally surround the hollow cylinders. Second sacrificial material is formed within upper regions of the hollow cylinders. At least some of the first sacrificial material of the first levels is replaced with conductive material. A planarized surface is formed to extend across an uppermost of the second levels, across the pillars and across the second sacrificial material. A third insulative material is formed over the planarized surface. Openings are formed to extend through the third insulative material to the second sacrificial material. The second sacrificial material is removed to extend the openings to upper surfaces of the second insulative material. Conductive plugs are formed within the extended openings. The conductive plugs comprise doped-semiconductor-material. Bitlines are formed to be coupled with the channel material of the pillars through the conductive plugs.
(91) In compliance with the statute, the subject matter disclosed herein has been described in language more or less specific as to structural and methodical features. It is to be understood, however, that the claims are not limited to the specific features shown and described, since the means herein disclosed comprise example embodiments. The claims are thus to be afforded full scope as literally worded, and to be appropriately interpreted in accordance with the doctrine of equivalents.

Claims

1. An integrated assembly, comprising: a channel-material-pillar extending vertically through a stack of alternating conductive levels and insulative levels; the channel-material-pillar comprising a first semiconductor material; the first semiconductor material extending below the alternating conductive levels and the insulative levels, the channel-material-pillar defining a hollow filled with insulative material; and a second semiconductor material directly against an upper region of the channel-material-pillar; the second semiconductor material having a higher dopant concentration than the first semiconductor material along an abrupt interfacial region.
2. The integrated assembly of claim 1 wherein the first and second semiconductor materials comprise a same semiconductor composition as one another.
3. The integrated assembly of claim 1 wherein the first and second semiconductor materials both

comprise silicon.
4. The integrated assembly of claim 1 wherein the second semiconductor material is in both of an amorphous phase and a polycrystalline phase.
5. The integrated assembly of claim 1 wherein one or more of the upper conductive levels of the stack are SGD levels, and wherein the interfacial region is above an uppermost of the SGD levels.
6. The integrated assembly of claim 1 wherein the first semiconductor-material is silicon having any dopant therein to a total dopant concentration of less than or equal to 1×10.sup.18 atoms/cm.sup.3.
7. The integrated assembly of claim 1 wherein the first semiconductor-material is silicon having any dopant therein to a total dopant concentration of less than or equal to 1×10.sup.16 atoms/cm.sup.3.
8. The integrated assembly of claim 1 wherein the second semiconductor-material is silicon having a total dopant concentration of greater than or equal to 1×10.sup.20 atoms/cm.sup.3.
9. The integrated assembly of claim 1 wherein the second semiconductor-material is silicon having a total dopant concentration of greater than or equal to 1×10.sup.21 atoms/cm.sup.3.
10. The integrated assembly of claim 1 wherein the second semiconductor material is in an amorphous phase.
11. The integrated assembly of claim 1 wherein the first semiconductor material comprises one or more of germanium, III/V semiconductor material and semiconductor oxide.
12. The integrated assembly of claim 1 wherein the first semiconductor material comprises one of the following dopant concentrations: less than about 10.sup.16 atoms/cm.sup.3; less than about 10.sup.15 atoms/cm.sup.3; or less than or equal to about 10.sup.18 atoms/cm.sup.3.
13. An integrated assembly, comprising: a channel-material-pillar extending vertically through a stack of alternating conductive levels and insulative levels; the channel-material-pillar comprising a first semiconductor material; and a second semiconductor material directly against an upper region of the channel-material-pillar; the first and second semiconductor materials being doped, the second semiconductor material having a higher dopant concentration than the first semiconductor material and joining to the first semiconductor along an abrupt interfacial region such that there is little to no mixing of dopant from the second semiconductor material into the first semiconductor material, wherein other than the second semiconductor material, only insulative material surrounds an upper region of the channel-material-pillar; and wherein the first semiconductor material comprises at least semiconductor oxide.
14. The integrated assembly of claim 13 wherein the first and second semiconductor materials comprise a same semiconductor composition as one another.
15. The integrated assembly of claim 13 wherein the first and second semiconductor materials both comprise silicon.
16. The integrated assembly of claim 13 wherein the second semiconductor material is in both of an amorphous phase and a polycrystalline phase.
17. The integrated assembly of claim 13 wherein one or more of the upper conductive levels of the stack are SGD levels, and wherein the interfacial region is above an uppermost of the SGD levels.
18. The integrated assembly of claim 13 wherein the first semiconductor-material is silicon having any dopant therein to a total dopant concentration of less than or equal to 1×10.sup.18 atoms/cm.sup.3.
19. The integrated assembly of claim 13 wherein the first semiconductor-material is silicon having any dopant therein to a total dopant concentration of less than or equal to 1×10.sup.16 atoms/cm.sup.3.
20. The integrated assembly of claim 13 wherein the second semiconductor material is in an amorphous phase.
21. The integrated assembly of claim 13 wherein the first semiconductor material comprises one of the following dopant concentrations: less than about 10.sup.16 atoms/cm.sup.3, less than about

10.sup.15 atoms/cm.sup.3: less than or equal to about 10.sup.18 atoms/cm.sup.3.
22. The integrated assembly of claim 13 wherein dopant from the second semiconductor material extends into the first semiconductor material to one of the following distances: less than or equal to about 10 nm into the first semiconductor material; less than or equal to about 5 nm into the first semiconductor material; or less than or equal to about 2 nm into the first semiconductor material.
23. An integrated assembly, comprising: a channel-material-pillar extending vertically through a stack of alternating conductive levels and insulative levels; the channel-material-pillar comprising a first semiconductor material; and a second semiconductor material directly against an upper region of the channel-material-pillar; the second semiconductor material having a higher dopant concentration than the first semiconductor material and joining to the first semiconductor along an abrupt interfacial region such that there is little to no mixing of dopant from the second semiconductor material into the first semiconductor material, wherein other than the second semiconductor material, only insulative material surrounds an upper region of the channel-material-pillar; and wherein the second semiconductor-material is silicon having a total dopant concentration of greater than or equal to 1×10.sup.20 atoms/cm.sup.3.
24. The integrated assembly of claim 23 wherein the second semiconductor-material has a total dopant concentration of greater than or equal to 1×10.sup.21 atoms/cm.sup.3.
25. The integrated assembly of claim 1 wherein the channel-material-pillar comprises a horizontal width that equals a horizontal width of the second semiconductor material.
26. The integrated assembly of claim 1 wherein, in a vertical cross section, the channel-material-pillar comprises opposite sidewalls collinear with opposite sidewalls of the second semiconductor material.
27. The integrated assembly of claim 1 further comprising a conductive interconnect over and against the second semiconductor material, the conductive interconnect comprising a trapezoid shape.
28. The integrated assembly of claim 1 further comprising a conductive interconnect over and against the second semiconductor material, the conductive interconnect comprising a metal-containing material.
29. The integrated assembly of claim 28 wherein the conductive interconnect comprises a first conductive interconnect, and further comprising a second conductive interconnect over and against the first conductive interconnect.
30. The integrated assembly of claim 1 further comprising: a conductive interconnect over the second semiconductor material; and a bit line over and against the conductive interconnect.
31. The integrated assembly of claim 13 wherein the channel-material-pillar comprises a horizontal width that equals a horizontal width of the second semiconductor material.
32. The integrated assembly of claim 13 wherein, in a vertical cross section, the channel-material-pillar comprises opposite sidewalls collinear with opposite sidewalls of the second semiconductor material.
33. The integrated assembly of claim 13 further comprising a conductive interconnect over and against the second semiconductor material, the conductive interconnect comprising a trapezoid shape.
34. The integrated assembly of claim 13 further comprising a conductive interconnect over and against the second semiconductor material, the conductive interconnect comprising a metal-containing material.
35. The integrated assembly of claim 34 wherein the conductive interconnect comprises a first conductive interconnect, and further comprising a second conductive interconnect over and against the first conductive interconnect.
36. The integrated assembly of claim 13 further comprising: a conductive interconnect over the second semiconductor material; and a bit line over and against the conductive interconnect.
37. The integrated assembly of claim 13 wherein the first semiconductor material extends below

the stack of the alternating conductive levels and the insulative levels.
38. The integrated assembly of claim 23 wherein the channel-material-pillar comprises a horizontal width that equals a horizontal width of the second semiconductor material.
39. The integrated assembly of claim 23 wherein, in a vertical cross section, the channel-material-pillar comprises opposite sidewalls collinear with opposite sidewalls of the second semiconductor material.
40. The integrated assembly of claim 23 further comprising a conductive interconnect over and against the second semiconductor material, the conductive interconnect comprising a trapezoid shape.
41. The integrated assembly of claim 23 further comprising a conductive interconnect over and against the second semiconductor material, the conductive interconnect comprising a metal-containing material.
42. The integrated assembly of claim 41 wherein the conductive interconnect comprises a first conductive interconnect, and further comprising a second conductive interconnect over and against the first conductive interconnect.
43. The integrated assembly of claim 23 wherein the first semiconductor material comprises a dopant concentration of less than about 10.sup.15 atoms/cm.sup.3.
44. The integrated assembly of claim 23 wherein the first semiconductor material comprises a dopant concentration of less than about 10.sup.16 atoms/cm.sup.3.
45. The integrated assembly of claim 1 wherein the dopant of the first semiconductor material comprises n-type dopant.
46. The integrated assembly of claim 1 wherein the dopant of the first semiconductor material comprises n-type dopant and p-type dopant.
47. The integrated assembly of claim 1 wherein the dopant of the second semiconductor material comprises n-type dopant.
48. The integrated assembly of claim 1 wherein the dopant of the second semiconductor material comprises n-type dopant and p-type dopant.
49. The integrated assembly of claim 13 wherein the dopant of the first semiconductor material comprises n-type dopant.
50. The integrated assembly of claim 13 wherein the dopant of the first semiconductor material comprises n-type dopant and p-type dopant.
51. The integrated assembly of claim 13 wherein the dopant of the second semiconductor material comprises n-type dopant.
52. The integrated assembly of claim 13 wherein the dopant of the second semiconductor material comprises n-type dopant and p-type dopant.
53. The integrated assembly of claim 23 wherein the dopant of the first semiconductor material comprises n-type dopant.
54. The integrated assembly of claim 23 wherein the dopant of the first semiconductor material comprises n-type dopant and p-type dopant.
55. The integrated assembly of claim 23 wherein the dopant of the second semiconductor material comprises n-type dopant.
56. The integrated assembly of claim 23 wherein the dopant of the second semiconductor material comprises n-type dopant and p-type dopant.